

400G PAM4 DCI Architecture

A Deep Dive into 3nm CMOS SerDes Design, High-Speed Clocking,
and Concatenated FEC Simulation

01. Lane Architecture

400G MONOLITHIC PHY OVERVIEW



ADC-Based SerDes

The architecture leverages a high-speed ADC-based receiver to handle severe Inter-Symbol Interference (ISI) in high-loss DCI channels. This allows for advanced DSP equalization.



4x106G Lane Mapping

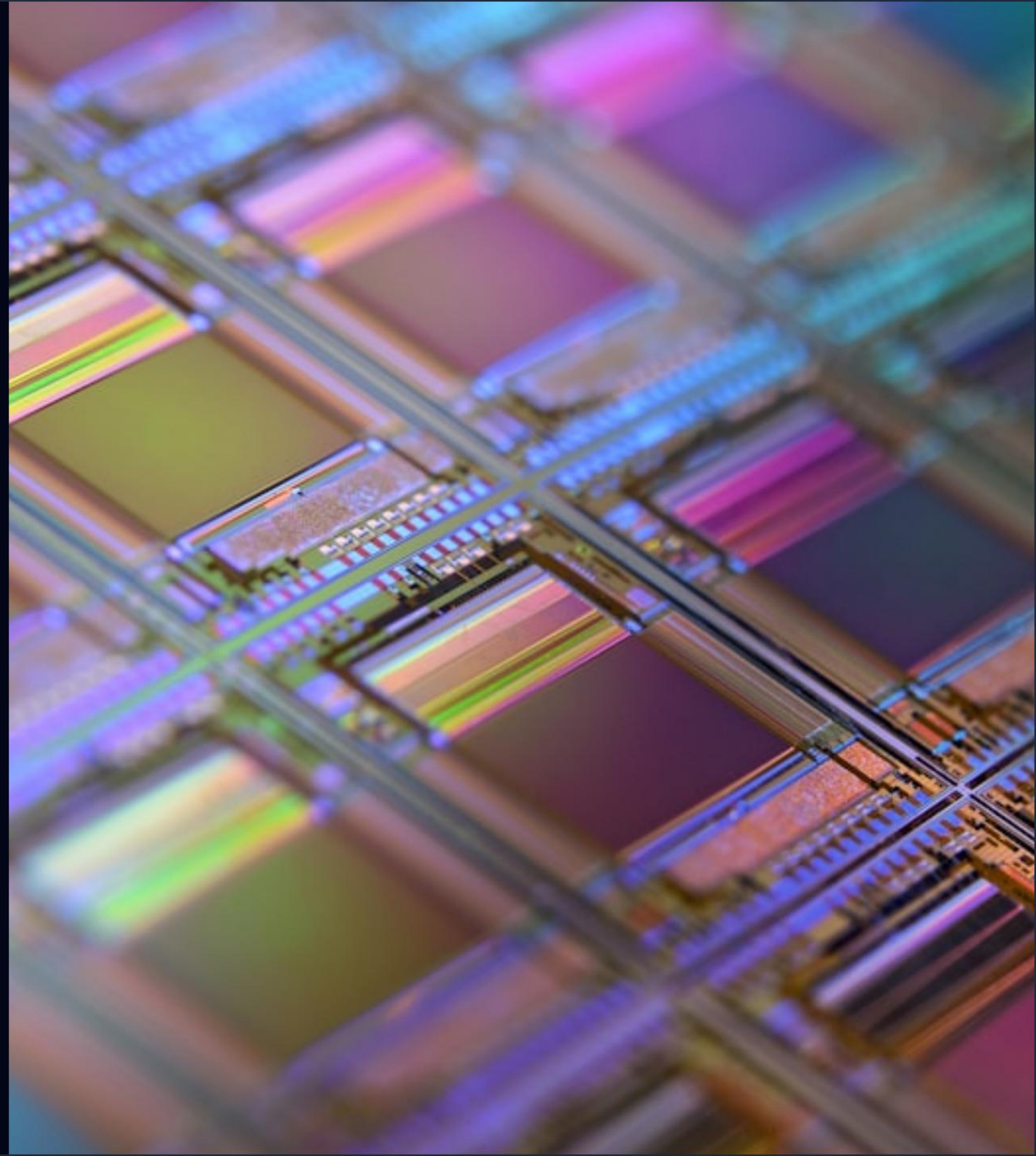
Conversion of eight system-side 53G lanes into four 106G (53 GBaud PAM4) line-side lanes, optimizing bandwidth density for QSFP-DD form factors.

3NM DIGITAL CORE

The 1.5625 GHz Clock Tree

In 3nm N3 nodes, wire resistance and voltage drop are primary constraints. A 1.5625 GHz internal digital clock drives a 64-bit wide backend bus to achieve 100 Gbps per lane throughput.

Advanced clock distribution networks utilize multi-stage buffering and shielding to mitigate intra-pair skew and maintain sub-100fs RJ.



PAM4 SI & CROSSTALK

- ⌋ **4-Lane Interference:** High-density packaging induces Near-End (NEXT) and Far-End (FEXT) crosstalk between adjacent differential pairs.
- 👁 **Eye Integrity:** PAM4 utilizes four levels (0, 1, 2, 3), reducing SNR margin by 9.5 dB compared to NRZ, making crosstalk mitigation critical.
- ⚡ **ISI Mitigation:** 31-tap FFE and 5-tap DFE used in 3nm DSP to resolve closed eyes in 30dB+ loss channels.

 PAM4 Eye Diagram

02. FEC Strategy

INNER CODE: LAYERED LDPC

40+

MBPS JAX THROUGHPUT

Soft-Decision Decoding

Inner Layered Belief Propagation (BP) decoder implemented in JAX/XLA. The model leverages **float16** state storage and **float32** compute to optimize RTX 3090 memory bandwidth.

$$Q_{v \rightarrow c} = L_v - R_{\text{old}, c \rightarrow v}$$

OUTER CODE: BCH CLEANUP

- 🔧 **Error Floor Mitigation:** Algebraic BCH decoding is used to clean up residual bit errors from LDPC "trapping sets" and "oscillations."
- ⚡ **Low Latency:** BCH decoders provide deterministic latency, critical for DCI links that cannot afford iterative decoding variance.
- 🛡️ **Trapping Set Escape:** Quantization in JAX (float16) introduces numerical "regularization" that helps the decoder escape sub-optimal local minima.

CONCATENATED PERFORMANCE



Net Coding Gain

The combination of LDPC(1944, 5/6) and BCH(2048, 1944) delivers >11 dB NCG at $1e-12$ BER floors.



JAX Scale-Up

Transitioning from 3090 single-GPU to A100 multi-GPU via `jax.pmap` allows simulation of $1e-15$ BER targets.



MATLAB Verification

Perfect alignment with MATLAB 'approxllr' math verifies the PRNG and AWGN channel noise floor.

03. Timing Recovery

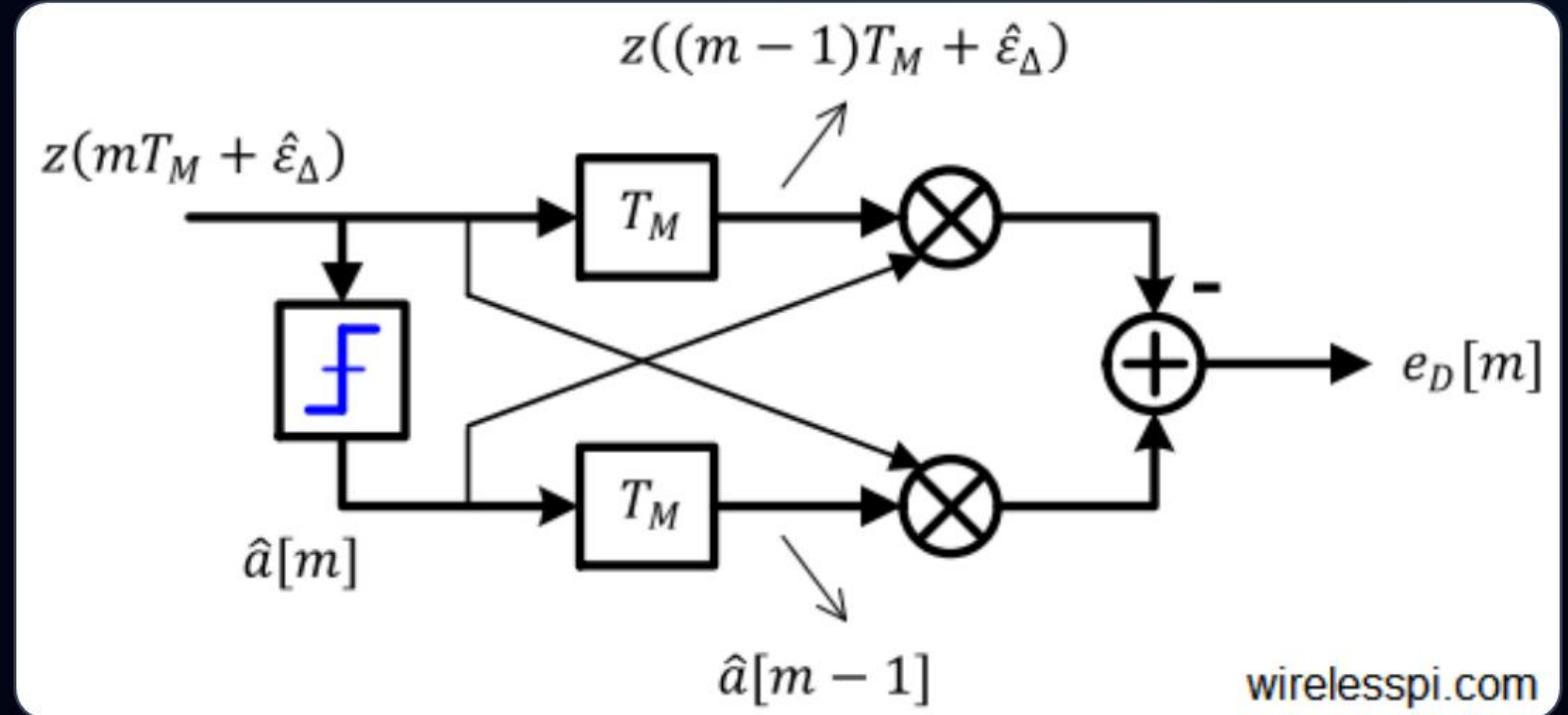
MUELLER-MÜLLER CDR

Baud-Rate Phase Detection

A Mueller-Müller detector operates at the baud rate (1x), significantly reducing ADC power consumption in 112G SerDes designs.

$$Z_k = X_k \hat{X}_{k-1} - X_{k-1} \hat{X}_k$$

The loop filters out jitter and tracks phase drift to maintain optimal sampling timing across 4 parallel lanes.



CONCLUSION & Q&A

3djax Project: GPU-Accelerated FEC Simulations.

Architecture: 3nm Monolithic 400G SerDes.

Innovation: JAX-based static graph optimization for high-speed hardware modeling.

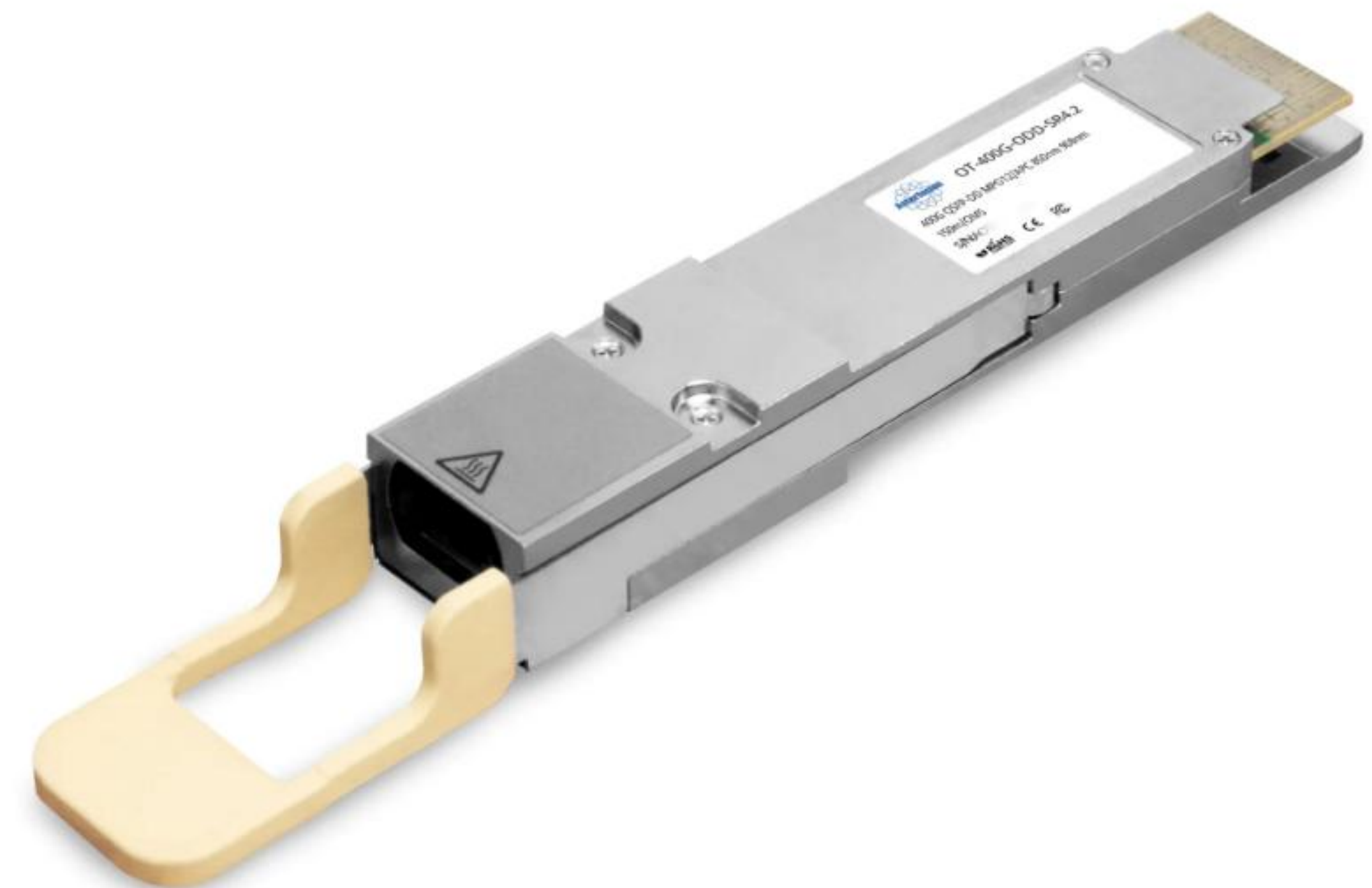
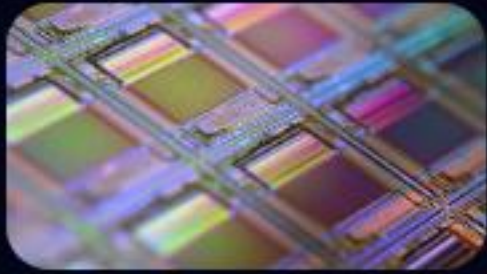
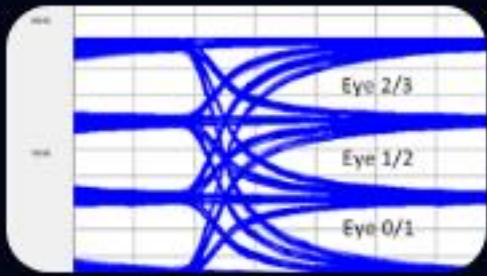


IMAGE SOURCES



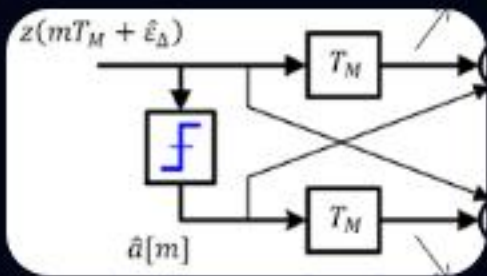
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Source: alliancechemical.com



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Source: helpfiles.keysight.com



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